

Weather and Climate Big Data

기상기후빅데이터

Hands-on Exercise 4: Time Series Modeling

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Today's Topics

- Time Series Data
- Time Series Modeling Approaches
- Hands-on: Temperature Prediction

ARIMA

- **Autoregressive Integrated Moving Average**
 - Used for time series forecasting of non-seasonal data
 - AR: Autogressive - uses past values
 - I: Integrated - differencing to remove trend
 - MA: Moving Average - uses past errors
- **Standard notation: ARIMA(p,d,q)**
 - AR(p): the number of lag observations
 - I(d): Degree of differencing
 - MA(q): the size of the moving average window

ARIMA

- **General form** (after differencing d times):

$$y'_t = \phi_1 y'_{t-1} + \cdots + \phi_p y'_{t-p} + \theta_1 \varepsilon_{t-1} + \cdots + \theta_q \varepsilon_{t-q} + \varepsilon_t$$

- **Example: ARIMA(3,1,2)**
 - First-difference of the data (d=1)
 - Uses 3 lags of past differenced values (AR terms)
 - And 2 lags of forecast errors (MA terms)

ARIMA

- **Best used when:**
 - Data shows trend but no seasonality
 - You want interpretable, lightweight models
 - Dataset is relatively small
- **For seasonal patterns, use:**
 - SARIMA (Seasonal ARIMA)
 - ARIMAX (Arima with exogenous variables)

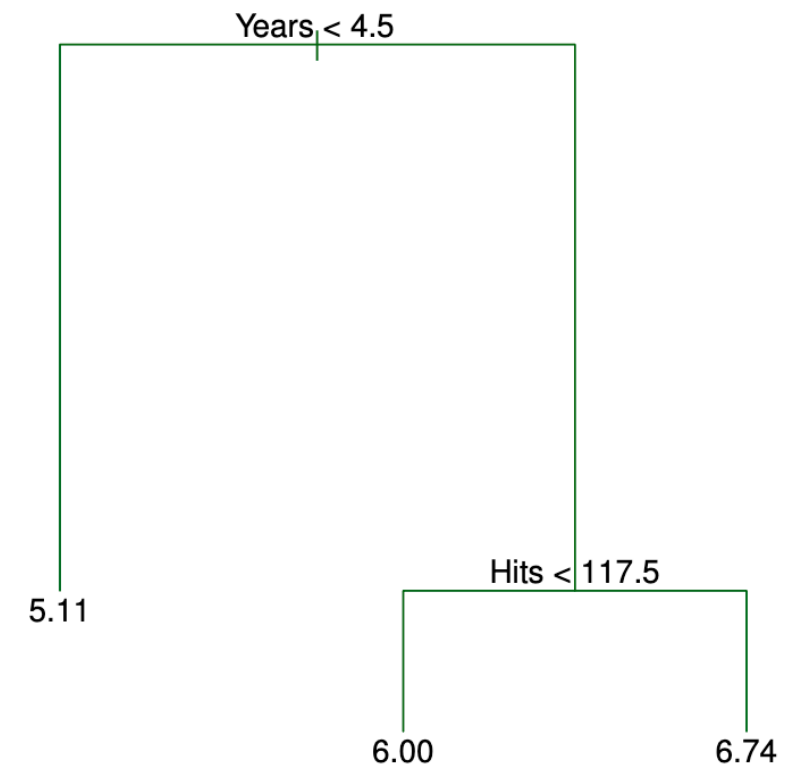
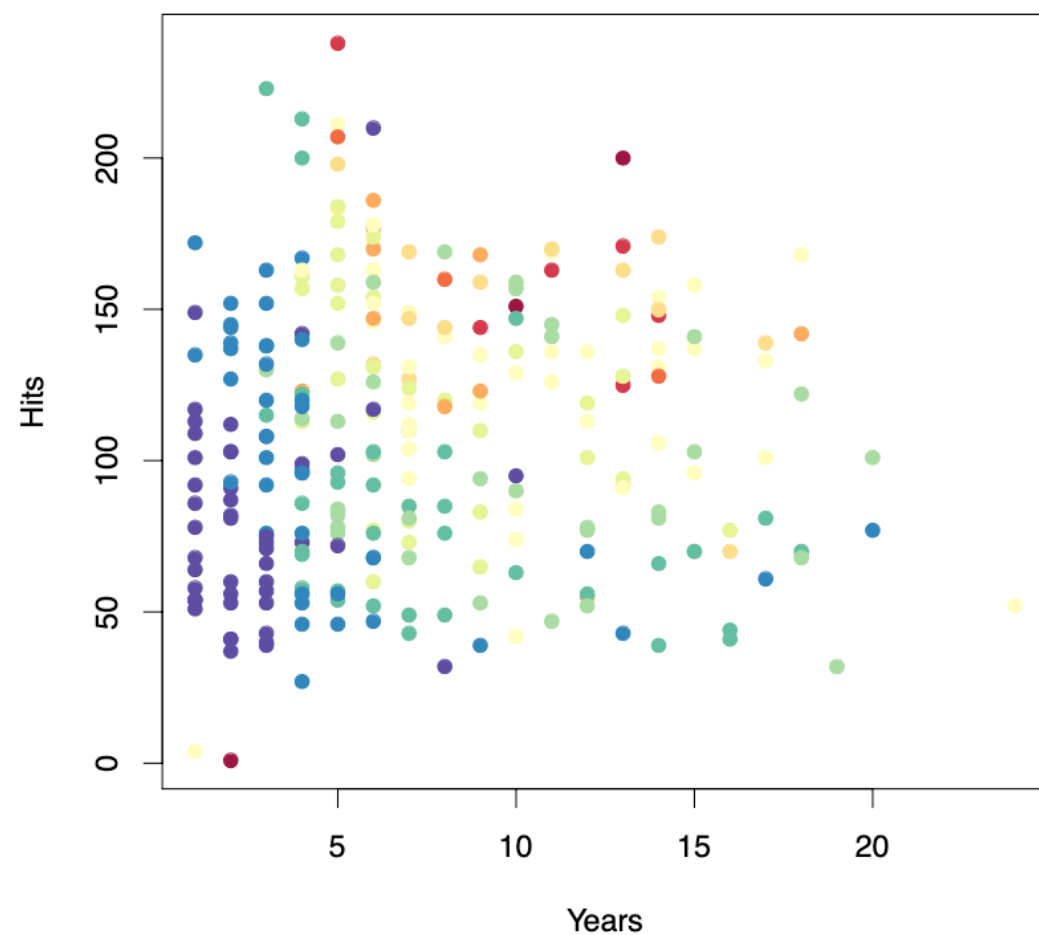
Python Code Example

```
from statsmodels.tsa.arima.model import ARIMA

model = ARIMA(y, order=(1, 1, 1))
model_fit = model.fit()
forecast = model_fit.forecast(steps=5)
```

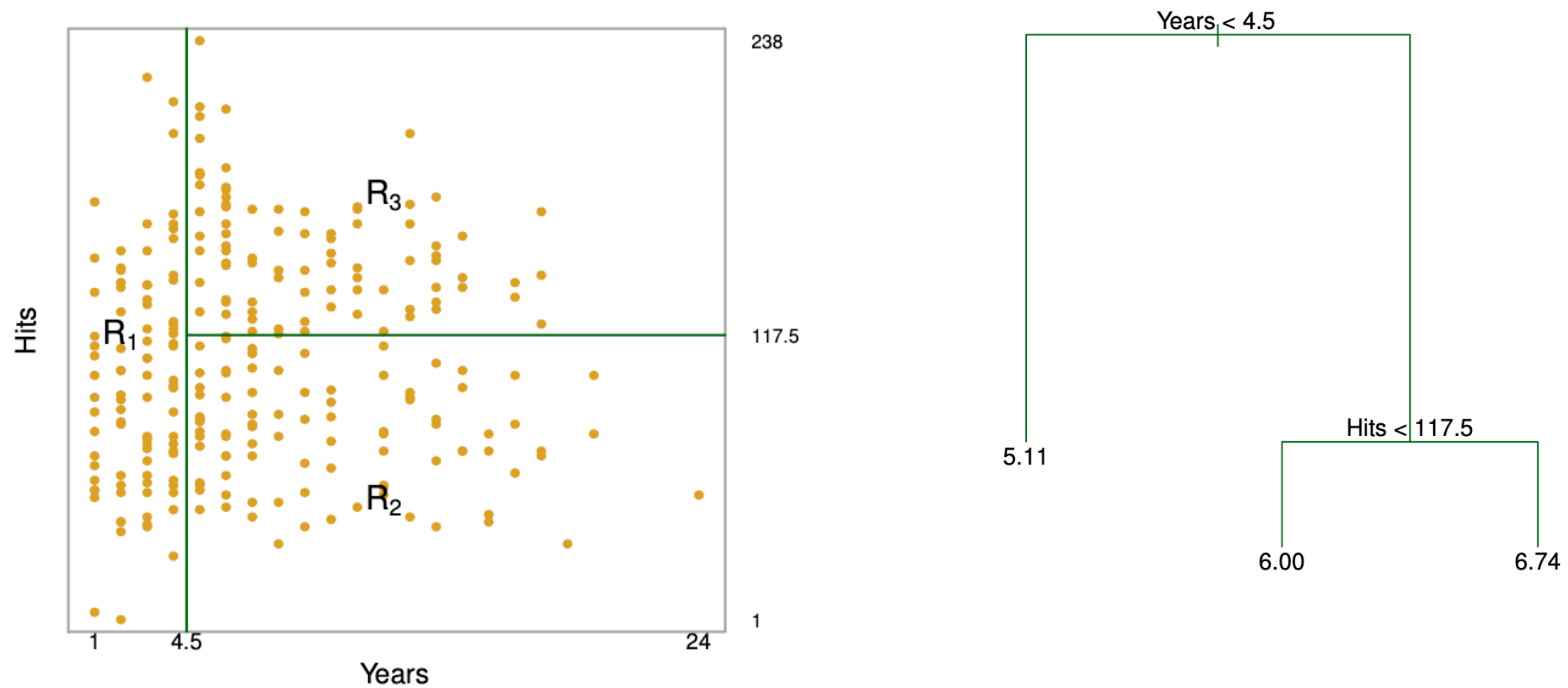
Random Forest

- An ensemble method based on decision trees



Random Forest

- An ensemble method based on decision trees



Random Forest

- Works for both classification and regression
- An ensemble method based on decision trees
- Combines multiple trees to improve prediction accuracy
- Uses bagging (bootstrap aggregating) + feature randomness
- Reduces overfitting compared to a single decision tree

Pros and Cons

- Advantages
 - Handles large datasets and high-dimensional features
 - Resistant to overfitting
 - No need to scale or normalize data
 - Works well even with missing values
- Limitations
 - Slower and heavier than a single tree
 - Less interpretable
 - Cannot model time/order dependencies

Python Code Example

```
from sklearn.ensemble import RandomForestRegressor

model = RandomForestRegressor(n_estimators=100)
model.fit(X_train, y_train)
preds = model.predict(X_test)
```

From RNNs to Attention

- Pre-2014: RNN, LSTM, GRU
 - RNNs struggled with long sequence data due to vanishing gradients
 - LSMT and GRU introduced to address memory limitations
- 2014: Attention Emerges
 - Proposed attention mechanism (Bahdanau, 2014)
 - Enabled better handling of long-term dependencies
- But still relied on RNN structures

From RNNs to Attention

colah's blog

Understanding LSTM Networks

Posted on August 27, 2015

Recurrent Neural Networks

Humans don't start their thinking from scratch every second. As you read this essay, you understand each word based on your understanding of previous words. You don't throw everything away and start thinking from scratch again. Your thoughts have persistence.

<https://colah.github.io/posts/2015-08-Understanding-LSTMs/>

From RNNs to Attention

NEURAL MACHINE TRANSLATION BY JOINTLY LEARNING TO ALIGN AND TRANSLATE

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ABSTRACT

Neural machine translation is a recently proposed approach to machine translation. Unlike the traditional statistical machine translation, the neural machine translation aims at building a single neural network that can be jointly tuned to maximize the translation performance. The models proposed recently for neural machine translation often belong to a family of encoder-decoders and encode a source sentence into a fixed-length vector from which a decoder generates a translation. In this paper, we conjecture that the use of a fixed-length vector is a bottleneck in improving the performance of this basic encoder-decoder architecture, and propose to extend this by allowing a model to automatically (soft-)search for parts of a source sentence that are relevant to predicting a target word, without having to form these parts as a hard segment explicitly. With this new approach, we achieve a translation performance comparable to the existing state-of-the-art phrase-based system on the task of English-to-French translation. Furthermore, qualitative analysis reveals that the (soft-)alignments found by the model agree well with our intuition.

by D Bahdanau · 2014 · Cited by 39170 -

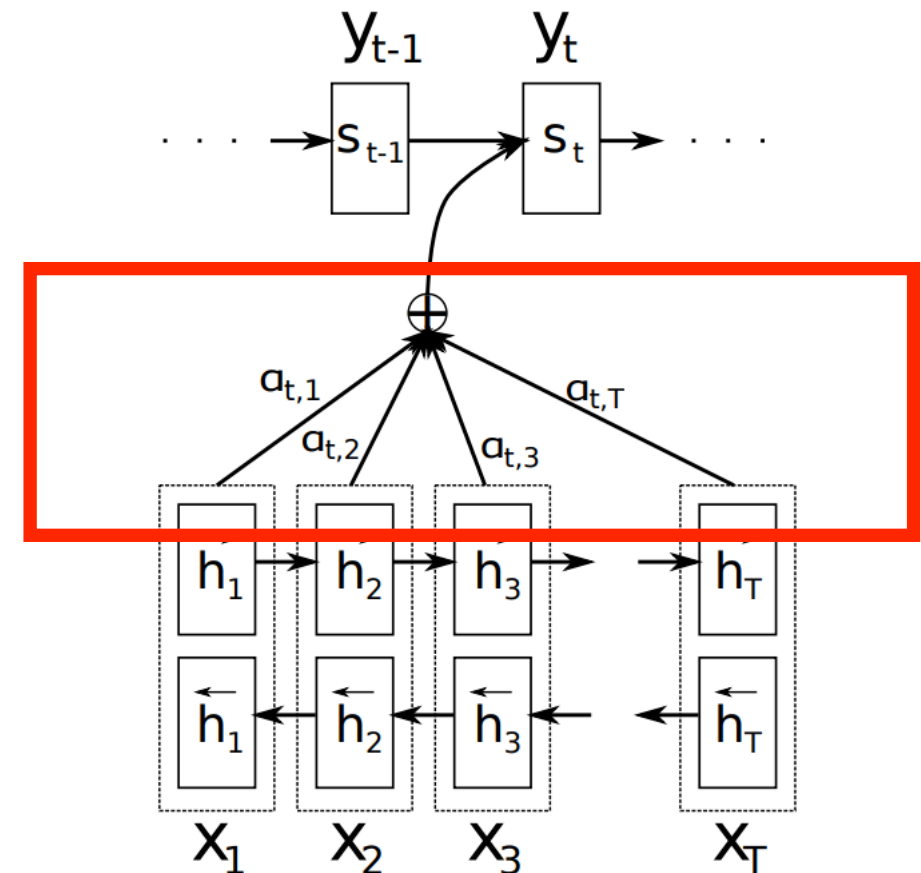


Figure 1: The graphical illustration of the proposed model trying to generate the t -th target word y_t given a source sentence $(x_1, x_2, \dots, x_T)$.

From RNNs to Attention

Attention Is All You Need

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Abstract

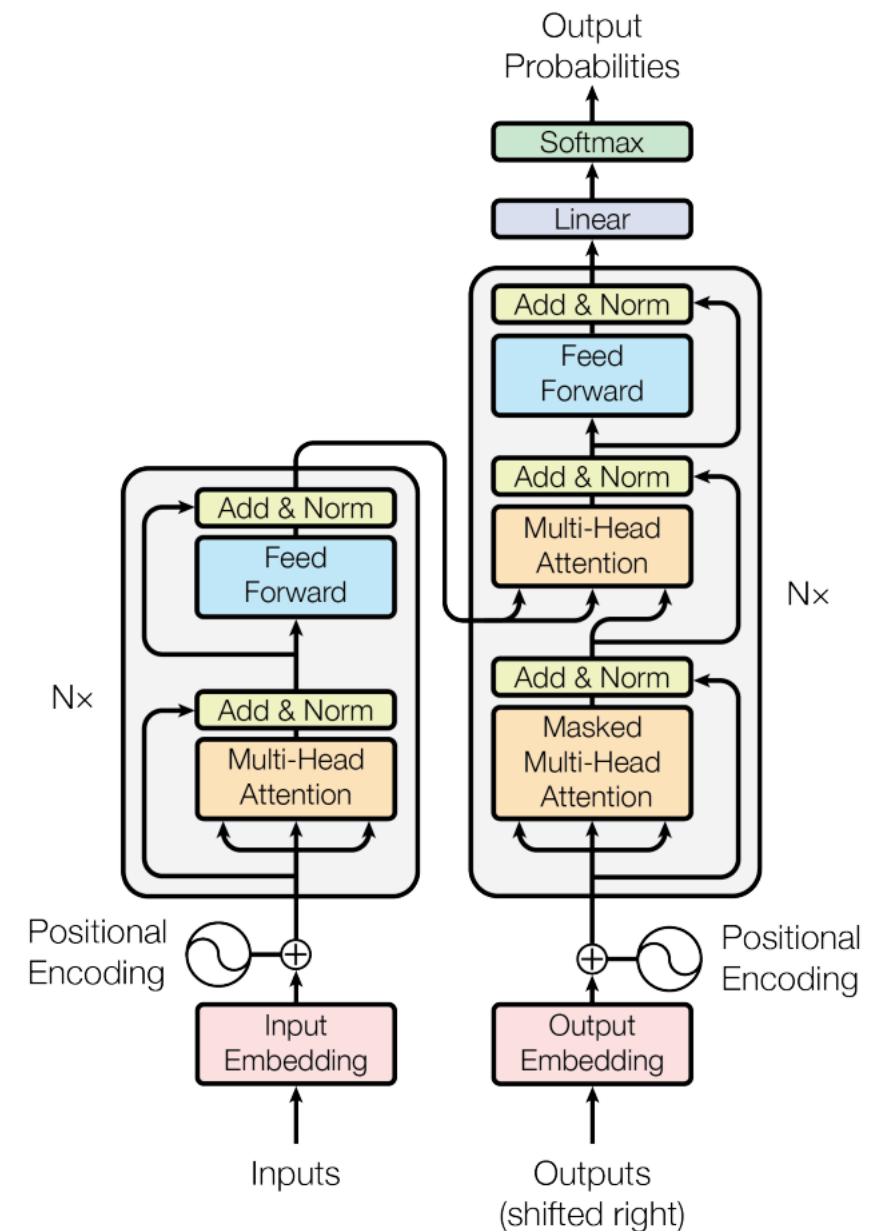


Figure 1: The Transformer - model architecture.

From RNNs to Attention



Journal of Hydrology

Volume 637, June 2024, 131389



Technical Note

Probing the limit of hydrologic with the Transformer network

Jiangtao Liu^a, Yuchen Bian^b, Kathryn Lawson^a, Chaopeng Shen^a

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<https://doi.org/10.1016/j.jhydrol.2024.131389> ↗

Highlights

- First time Transformer achieved the same performance as LSTM on CAMELS dataset.
- Unmodified vanilla Transformer could not reach LSTM's performance.
- A non-recurrent connection was added to the Transformer, supporting parallelism.
- Transformers may have scale advantages for larger datasets.
- LSTMs and Transformers are likely nearing the prediction limits of the dataset.

From RNNs to Attention

OpenReview.net

[← Go to ICLR 2024 Conference homepage](#)

Mamba: Linear-Time Sequence Modeling with Selective State Spaces

Albert Gu, Tri Dao

22 Sept 2023 (modified: 11 Feb 2024) Submitted to ICLR 2024 Everyone Revisions BibTeX

Primary Area: representation learning for computer vision, audio, language, and other modalities

Code Of Ethics: I acknowledge that I and all co-authors of this work have read and commit to adhering to the ICLR Code of Ethics.

Keywords: Sequence model, language model, state space model, RNN, SSM, S4, Mamba

Submission Guidelines: I certify that this submission complies with the submission instructions as described on <https://iclr.cc/Conferences>.

TL;DR: We introduce a selection mechanism into state space models, leading to state-of-the-art results on general sequence modeling including

Abstract:
Foundation models, now powering most of the exciting applications in deep learning, are almost universally based on the Transformer architecture's self-attention module. Many subquadratic-time architectures such as linear attention, gated convolution and recurrent models, and structured state space models have been developed to address Transformers' computational inefficiency on long sequences, but they have not performed as well as attent

PatchTST: Main Concepts

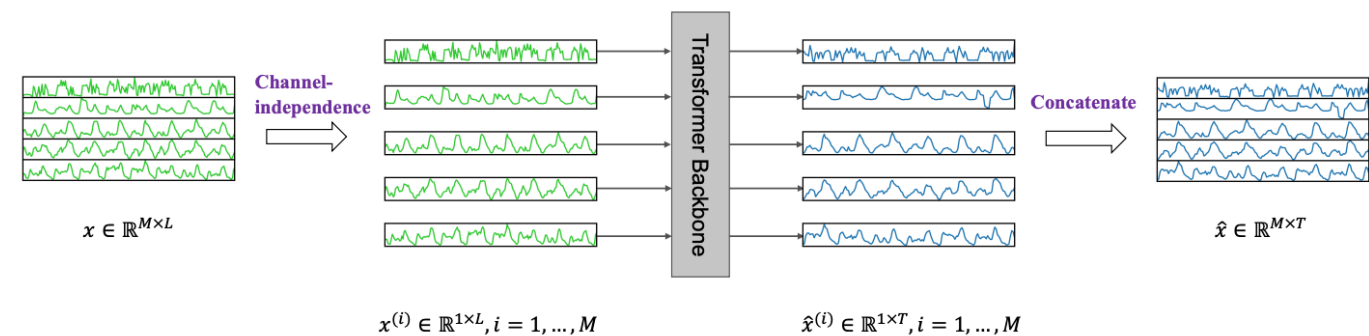
- Designed specifically for long-horizon time series forecasting
- Inspired by Vision Transformers (ViT)
- Patchify: split time series into fixed-length segments (patches)
- Each patch acts as a token (like words in NLP)
- Enables learning of local patterns and reduces sequence length

PatchTST: Main Concepts

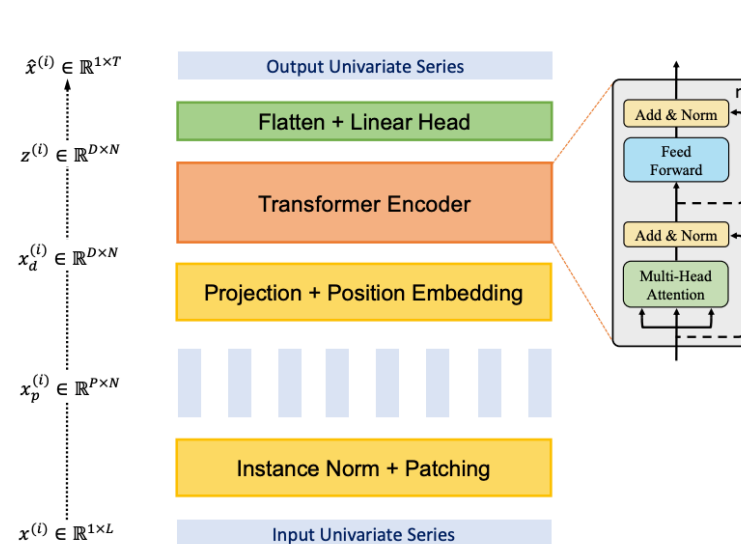
Published as a conference paper at ICLR 2023

A TIME SERIES IS WORTH
LONG-TERM FORECASTING

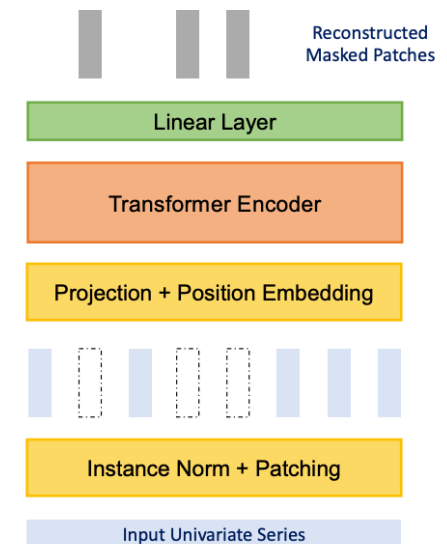
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(a) PatchTST Model Overview



(b) Transformer Backbone (Supervised)



(c) Transformer Backbone (Self-supervised)

<https://arxiv.org/pdf/2211.14730>

Extended Readings

- <https://jinglescode.github.io/2020/05/27/illustrated-guide-transformer/>
- <https://towardsdatascience.com/transformers-141e32e69591>
- <https://jalammar.github.io/illustrated-transformer/>
- <https://medium.com/analytics-vidhya/neural-machine-translation-using-bahdanau-attention-mechanism-d496c9be30c3>